

ECMO for Meconium Aspiration Syndrome: Support for Relaxed Entry Criteria

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We compared the morbidity of patients with meconium aspiration syndrome (MAS) with that in patients with all other respiratory conditions treated with extracorporeal membrane oxygenation (ECMO) (no MAS). If ECMO for MAS was associated with a lower complication rate, then relaxed ECMO entry criteria could be considered. A retrospective review was performed of all patients in the national extracorporeal life support (ELSO) registry from 1989 to 2004. Complications were divided into mechanical, hematologic, neurologic, renal, pulmonary, cardiovascular, infectious, and metabolic categories. MAS and no-MAS patients were divided into venovenous (VV) or veno-arterial (VA) ECMO categories, based on mode of ECMO used, and number of complications per patient in each category was determined. Statistical significance was determined by Chi-square test. A total of 1587 patients (700 MAS, 887 no MAS) on VV ECMO and 2723 (572 MAS, 2151 no MAS) on VA ECMO were identified with a total of 2415 complications in MAS and 9550 in no-MAS patients. Overall, MAS patients had a significantly lower number of complications per patient in each category versus no-MAS patients. These results indicate that regardless of type of ECMO, there are fewer complications on ECMO in MAS versus no-MAS patients. These data support the consideration of relaxed ECMO entry criteria for MAS. *ASAIO Journal* 2007; 53:489–491.

Meconium aspiration syndrome (MAS) is a common cause of neonatal respiratory failure, occurring in nearly 5% of live births,¹ and is the most frequent indication for initiation of extracorporeal membrane oxygenation (ECMO).² Survival rates for neonates with MAS placed on ECMO are at 94%.³ Furthermore, delayed initiation of ECMO has been shown to increase mortality rates, ventilator duration, and hospital stay.³ Despite these data, many practitioners delay initiation of ECMO because of the considerable overall morbidity observed with ECMO.

Currently, the entry criteria for neonatal ECMO for respiratory failure are the same regardless of the underlying cause. However, the rates of survival after ECMO vary widely, de-

pending on the etiology of respiratory distress. For instance, whereas the survival for MAS is at 94%, the survival for congenital diaphragmatic hernia is near 53%.⁴ These data together suggest that MAS patients may benefit from earlier implementation of ECMO and that uniform criteria for ECMO usage may delay its use in a subset of patients that may benefit.

We compared the morbidity of patients with MAS with that in patients with all other respiratory conditions treated with ECMO (no MAS). If ECMO for MAS was associated with a lower complication rate, then relaxed ECMO entry criteria for patients with MAS could be considered.

Methods

The international ECMO registry of the Extracorporeal Life Support Organization (ELSO, Ann Arbor, MI) contains voluntarily reported data from more than 145 centers worldwide. The data include patient demographics, clinical course, and clinical outcomes information for all patients treated with ECMO at each of these centers.

The registry was queried for all neonatal patients with respiratory failure treated with both venovenous (VV) and veno-arterial (VA) ECMO from 1989 to 2004. These patients were further subdivided into patients with respiratory failure from MAS and from all other causes (no MAS). Within these populations, mechanical, hematologic, neurologic, renal, pulmonary, cardiovascular, infectious, and metabolic complications were identified.

Mechanical complications involved oxygenator failure; raceway rupture or other tubing rupture; pump or heat exchanger malfunction; clots in the oxygenator, bridge, bladder, hemofilter; air in the circuit; cracks in the connectors; and problems with the cannulas. Hematologic complications included gastrointestinal, cannulation, or surgical site bleeding; disseminated intravascular coagulation (DIC); and hemolysis (plasma Hgb >50 mg/dL). Neurologic complications included clinical brain death, clinical seizures, EEG seizures, and infarction or hemorrhage (seen on ultrasound or computed tomography). Renal complications included creatinine >1.5, hemodialysis, and hemofiltration. Pulmonary complications included pneumothorax and pulmonary hemorrhage that required intervention. Cardiovascular complications included inotropes on ECMO, cardiopulmonary resuscitation required, myocardial stun by echocardiography, cardiac arrhythmias, hypertension requiring vasodilator treatment, PDA, and tamponade requiring intervention. Infectious complications included culture-proven new infections or white blood cell count <1500. Finally, metabolic complications included glucose <40 or >240, pH <7.20 or >7.60, and

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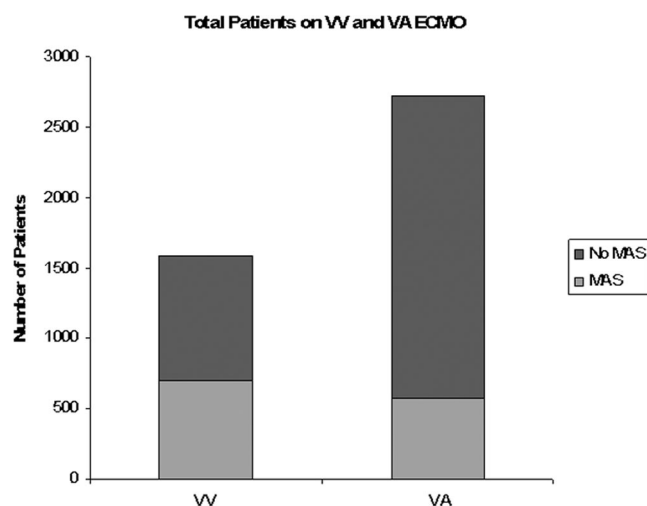


Figure 1. Total number of patients on VV and VA ECMO (MAS, meconium aspiration syndrome; no MAS, all other respiratory conditions treated with ECMO; VV, veno-venous ECMO; VA, veno-arterial ECMO).

hyperbilirubinemia (>2 direct, >13 indirect, or >15 total). These complications were classified as ECMO related and were entered into the ELSO registry if they occurred while the patient was on ECMO.

Statistical analysis was performed with the use of the χ^2 test with a significant difference set at $p < 0.05$. Data are expressed as mean $\pm$ standard error.

Results

Total Number of Patients With Respiratory Failure: VV and VA ECMO

The data depicted in **Figure 1** show the total number of patients treated on VV and VA ECMO. There were 1587 patients (700 MAS, 887 no MAS) on VV ECMO and 2723 patients (572 MAS, 2151 no MAS) on VA ECMO. **Figure 2** shows the percentage of MAS patients treated with VV ECMO

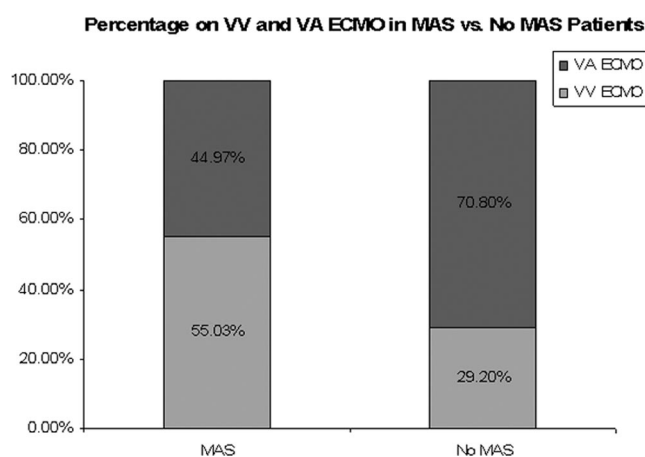


Figure 2. Percentage on VV and VA ECMO in MAS versus no-MAS patients (MAS, meconium aspiration syndrome; no MAS, all other respiratory conditions treated with ECMO; VV, veno-venous ECMO; VA, veno-arterial ECMO).

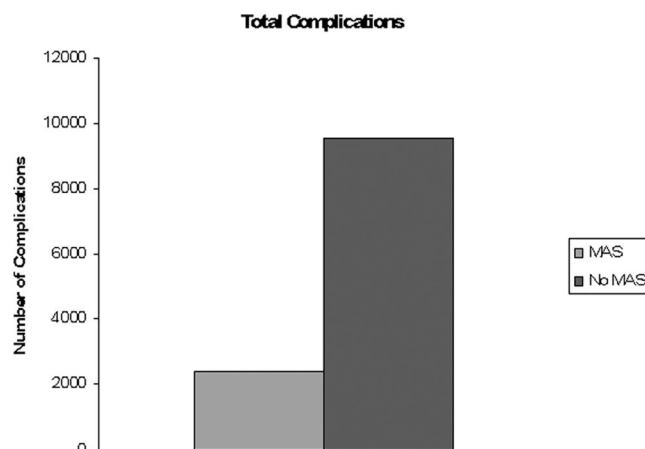


Figure 3. Total number of complications in each patient group (MAS, meconium aspiration syndrome; no MAS, all other respiratory conditions treated with ECMO).

(55.03%) and VA ECMO (44.97%) and the percentage of no MAS treated with VV ECMO (29.20%) and VA ECMO (70.80%).

Complications in MAS and No MAS Groups

The data depicted in **Figure 3** show that there were 11,965 total complications (2415 MAS and 9550 no MAS) and a total of 4310 patients with an overall rate of 2.78 complications per patient. When divided into MAS and no-MAS groups, there were 1.90 complications per MAS patient and 3.14 complications per no-MAS patient.

Data in **Table 1** show a significantly higher number of complications per patient in all categories except pulmonary and infectious as follows: Mechanical: VV no-MAS group (0.757 ± 0.042) vs. the VV MAS group (0.501 ± 0.036); VA no-MAS group (1.038 ± 0.031) vs. the VA MAS group (0.647 ± 0.051); hematologic: VV no-MAS group (0.233 ± 0.018) vs. the VV MAS group (0.124 ± 0.014); VA no-MAS group (0.339 ± 0.013) vs. the VA MAS group (0.227 ± 0.022); neurologic: VV no-MAS group (0.143 ± 0.014) vs. the VV MAS group (0.079 ± 0.012); VA no-MAS group (0.266 ± 0.012) vs. the VA MAS group (0.198 ± 0.022); renal: VV no-MAS group (0.289 ± 0.019) vs. the VV MAS group (0.119 ± 0.014); VA no-MAS group (0.377 ± 0.014) vs. the VA MAS group (0.205 ± 0.020); pulmonary: VV no-MAS group (0.095 ± 0.010) vs. the VV MAS group (0.083 ± 0.011); VA no-MAS group (0.156 ± 0.084) vs. the VA MAS group (0.128 ± 0.015); cardiovascular: VV no-MAS group (0.754 ± 0.027) vs. the VV MAS group (0.529 ± 0.027); VA no-MAS group (0.891 ± 0.019) vs. the VA MAS group (0.664 ± 0.031); infectious: VV no-MAS group (0.047 ± 0.007) vs. the VV MAS group (0.034 ± 0.007); VA no-MAS group (0.090 ± 0.007) vs. the VA MAS group (0.059 ± 0.010); metabolic: VV no-MAS group (0.171 ± 0.016) vs. the VV MAS group (0.094 ± 0.014); VA no-MAS group (0.255 ± 0.012) vs. the VA MAS group (0.182 ± 0.020).

Discussion

We have shown that MAS patients have a significantly lower incidence of mechanical, hematologic, neurologic, renal, car-

Table 1. Complications While on ECMO in MAS and No-MAS Patients

Category	MAS (VV)	No MAS (VV)	MAS (VA)	No MAS (VA)	Units
Mechanical	0.50 ± 0.04	0.76 ± 0.04*	0.65 ± 0.05	1.04 ± 0.03†	Complication/Pt.
Hematologic	0.12 ± 0.01	0.23 ± 0.02*	0.23 ± 0.02	0.34 ± 0.01†	Complication/Pt.
Neurologic	0.08 ± 0.001	0.14 ± 0.01*	0.20 ± 0.02	0.27 ± 0.01†	Complication/Pt.
Renal	0.12 ± 0.01	0.29 ± 0.02*	0.21 ± 0.02	0.38 ± 0.01†	Complication/Pt.
Pulmonary	0.08 ± 0.01	0.10 ± 0.01	0.13 ± 0.02	0.16 ± 0.08	Complication/Pt.
Cardiovascular	0.53 ± 0.03	0.75 ± 0.03*	0.66 ± 0.03	0.89 ± 0.02†	Complication/Pt.
Infectious	0.03 ± 0.01	0.05 ± 0.01	0.06 ± 0.01	0.09 ± 0.01	Complication/Pt.
Metabolic	0.09 ± 0.01	0.17 ± 0.02*	0.18 ± 0.02	0.26 ± 0.01†	Complication/Pt.

* Indicates statistical significance, $p < 0.05$ compared with MAS (VV).

† Indicates statistical significance, $p < 0.05$ compared with MAS (VA).

MAS, Meconium aspiration syndrome; no MAS, all other respiratory conditions treated with ECMO; VV, veno-venous ECMO; VA, veno-arterial ECMO.

diovascular, and metabolic complications per patient compared with their no-MAS counterparts. In addition, this difference was seen regardless of type of ECMO used. Finally, MAS patients are treated with VV ECMO more frequently than their no-MAS counterparts.

Utilization of adjuvant therapies such as high-frequency oscillatory ventilation (HFOV) and inhaled nitric oxide (iNO) has been advocated as an alternative to ECMO therapy.^{5,6} In addition, use of HFOV and iNO have been shown to improve oxygenation compared with standard ventilation.⁷ However, a prospective clinical trial examining the efficacy of HFOV versus conventional ventilation found that there was no difference in need for ECMO, treatment failure, complications, or outcome.⁸ These data imply that although oxygenation may be improved with adjuvant therapies such as HFOV and iNO, a significant number of patients (>40%) still experience treatment failure.⁸

In this large segment of patients, trials of ventilation (either standard, HFOV and/or iNO) do not decrease the need for ECMO and potentially delay initiation of ECMO. Gill *et al.* have demonstrated that delay in initiation of ECMO in patients with MAS leads to increased mortality rates, increased time on ECMO, increased time on the ventilator after ECMO, and increased hospital stay. In addition, prolonged trials of HFOV or iNO can lead to increased time on ventilator after ECMO.³

Of the causes for neonatal respiratory failure requiring ECMO, MAS has the highest survival rate,⁹ indicating that treatment of MAS with ECMO has a high success rate. In addition, more patients with MAS received VV ECMO than their no-MAS counterparts. Since this method of ECMO requires only internal jugular cannulation without carotid ligation, there is less morbidity required in placing the patient on ECMO. Taken with our findings that MAS patients have fewer complications, there is evidence that MAS may be a unique clinical entity that should be considered separately from other forms of neonatal respiratory distress. We offer that MAS be considered separately from other forms of neonatal respiratory failure. In addition, the entry criteria for ECMO should be relaxed for MAS to allow these patients earlier access to ECMO. A fundamental limitation of this study is that we do not have morbidity data for patients with MAS who have severe respiratory failure but do not meet ECMO criteria (OI = 20 to 40). The comparison of morbidity for this group to those

patients treated with ECMO would give more direct information regarding the relative morbidity of either treatment modality.

Treatment of MAS with ECMO has a higher degree of success and a lower incidence of complications than other forms of neonatal respiratory failure. In addition, lower morbidity VV ECMO is used more often than VA ECMO in MAS versus no-MAS patients. Therefore, ECMO is a less morbid therapy for MAS patients, and relaxed ECMO entry criteria would allow more MAS patients to be placed on ECMO earlier, potentially leading to improved patient outcome.

Acknowledgments

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